

Photon-Energy Transfer and Companion Deposition

A phenomenological framework for redshift, additional gravity and reservoir emission in a nonexpanding universe

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Abstract

We assess a hypothetical nonexpanding-universe framework in which photons transfer energy into traveling companions and some energy enters gravitationally active reservoirs. The empirical exact-third retention law remains the reference: galaxy rotation predictions improve over ordinary matter alone but remain worse in aggregate than the tested MOND relation; lensing and absolute supply remain unresolved. Updated calculations enforce reciprocal loading, finite storage, outgoing-wave feedback and energy bookkeeping. Collective enhancement can accelerate transfer but does not establish permanent retention. A narrow-angle Gaussian interaction cannot preserve an unchanged long-path transfer rate. A known threshold-mixture identity reproduces the reference curve by inverse design; finite thresholds closely preserve existing galaxy predictions, while assumed formation and release kinetics expose substantial history dependence. These are conditional mechanisms and numerical consistency results, not a unified microscopic derivation or new independent observations. No branch yet explains redshift, event durations, gravity, supply and persistence together.

1. Motivation and scientific scope

The objective is to ask whether energy originating in radiation can account for some phenomena conventionally attributed to additional gravitating matter, while a related propagation process accounts for redshift without metric expansion. The nonexpanding universe is a stipulated setting of the model, not an observational conclusion of this paper. Observed spectra, stellar motions, lensing images and radiation remain constraints to explain. No particular universe age or total size is imposed.

The absence of an identified dark-matter particle motivates alternatives but is not evidence for this specific mechanism. For example, the LZ collaboration's December 2025 analysis reported no WIMP signal in its stated search range [1]. That result constrains particle properties rather than erasing gravitational evidence. Our companions are likewise unobserved. Calling them radiation-derived rather than primordial does not exempt them from the evidential burden placed on an additional gravitating component.

We distinguish three comparisons: ordinary-matter Newtonian dynamics as a control; specified MOND and dark-halo models as gravitational competitors; and expanding cosmology as a separate propagation and large-scale benchmark. A successful galaxy fit alone cannot establish superiority over all three. Published distances may be adopted as scenario inputs, but their calibration and any dependence on the competing model must be recorded. A distance inferred from the same redshift law under test cannot serve as independent validation of that law.

2. Core postulates and unresolved choices

- P1. Photons progressively transfer energy into a companion sector. A surviving photon can have lower energy; simple removal of whole photons is insufficient.
- P2. Traveling companions are taken to propagate at light speed and retain their energy until an interaction. Their microscopic identity, polarization and relation to gravitons remain unspecified.
- P3. Capture transfers energy and momentum into a bound reservoir. The reservoir adds gravity under the reference effective-density prescription. Ordinary matter continues to gravitate.
- P4. The calibrated retention fraction has an exponent fixed to one-third. It is an empirical organizing rule, not a derived fundamental constant.
- P5. A captured reservoir may redistribute internally. Movement to deeper binding releases energy that must be accounted for together with support, heat and outgoing flux.
- P6. Companion-powered emission in black-hole environments is an optional branch. Evaporation of stored rest energy and release of settling energy are distinct mechanisms and must not be counted together as the same available budget.

An environmental time field remains a possible cause of P1, but no unique environmental clock law is established. Low gravity, absolute potential, density, curvature and tidal field are different candidate environmental variables. Substituting one for another changes predictions and requires an explicit choice. A freely chosen time coordinate by itself is not an observable interaction.

3. Minimal equations and provenance

3.1 Photon energy and redshift

Assume a local fractional energy-loss coefficient α per unit path length. With the standard photon energy-frequency relation, the deterministic reference law is

$$\frac{dE_\gamma}{d\ell} = -\alpha E_\gamma, \quad 1 + z = \frac{E_{\text{emit}}}{E_{\text{obs}}} = \exp\left(\int \alpha d\ell\right). \quad (1)$$

Provenance: the exponential solution is known mathematics and appears in other redshift models. Our proposed novelty would be an independently predictive physical equation for α , not equation (1). The archived illustrative constant is 0.0002488993 per Mpc; it does not establish broad-spectrum or independent distance-redshift agreement. At 100 million light-years it gives z approximately 0.00766 for the propagation component alone. Source/observer motions and ordinary endpoint gravitational effects must be treated separately.

An arrival-time map is also required:

$$t_o = F(t_e, \text{path}), \quad \frac{\Delta t_o}{\Delta t_e} \approx \frac{\partial F}{\partial t_e}. \quad (2)$$

Provenance: a general timing identity. Equation (1) does not determine equation (2). A static delay common to both ends of an event does not stretch its duration. The observed supernova width-redshift relationship [2] must be reproduced without using it merely to assign a new arbitrary factor. Different responses of electromagnetic and gravitational signals must also meet the GW170817 timing constraint [3], including

source emission-delay uncertainty.

3.2 Transport and capture

For energy densities u and energy fluxes F , a minimal conversion/capture ledger is

$$\partial_t u_Y + \nabla \cdot \mathbf{F}_Y = j_* - Q_{Yc}, \quad \partial_t u_c + \nabla \cdot \mathbf{F}_c = Q_{Yc} - C. \quad (3)$$

$$\partial_t u_d + \nabla \cdot \mathbf{F}_d = C - S_d. \quad (4)$$

Provenance: standard continuity equations; identifying the exchange terms with companions is proposed. S_d represents an actual energy transfer out of the deposited component and must enter another component or a boundary flux. If settling does gravitational work, kinetic and field-energy terms must be included; these three equations alone are not the full ledger. In a relativistic completion the total stress-energy exchanges must balance. This is not a claim that general relativity has a unique local gravitational energy density.

3.3 Retention and the reference deposited profile

$$X = \frac{L_{3.6}/(10^9 L_\odot)}{(R_d/\text{kpc})^2}, \quad \eta(X) = \frac{X^{1/3}}{1 + X^{1/3}}. \quad (5)$$

Provenance: the Hill-function form is established. Its radiation-density proxy, exponent choice and capture interpretation are project hypotheses. Here L refers to the adopted 3.6-micron luminosity proxy, not a measured total historical incident energy.

$$a = q_a R_d, \quad \kappa(r) = \frac{k_0}{[1 + (r/a)^2]^2}, \quad J(r) = \frac{1}{2} \int_{-1}^1 e^{-\tau(r, \mu)} d\mu. \quad (6)$$

The optical depth tau is the integral of kappa along the incoming ray. The effective stored density is

$$\rho_d(r) = \frac{2C_0 \eta(X) J(r)}{[1 + (r/a)^2]^2}, \quad M_d(< r) = 4\pi \int_0^r \rho_d(x) x^2 dx. \quad (7)$$

Provenance: ray attenuation, angular averaging and spherical integration are known. The opacity shape and its stored-density interpretation are proposed. The current fitted constants are $C_0=4.72858924e7$ solar masses/kpc cubed, $k_0=0.2039029004/\text{kpc}$ and $q_a=2.770766589$. C_0 is accumulated effective normalization, not a measured companion supply or universe age. The capture radius factor q_a differs from the contraction factor s below.

$$v_c^2(r) = v_b^2(r) + \frac{GM_d(< r)}{r}. \quad (8)$$

Provenance: standard Newtonian spherical-source gravity with the proposed effective mass inserted. Equating a deposited energy density to mass density via energy divided by c squared is only a nonrelativistic reference closure. Traveling radiation, pressure and field stress require a more complete treatment. No adjustable energy-to-gravity amplification is derived here.

4. Optional physical branches retained for investigation

4.1 Phase-dependent settling and its energy outlet

The earlier phase-selector pilot uses an ideal homogeneous Bose-gas criterion:

$$T = \frac{m\sigma^2}{k_B}, \quad T_c = \frac{2\pi\hbar^2}{mk_B} \left[\frac{\rho_d/m}{\zeta(3/2)} \right]^{2/3}, \quad f = \max[0, 1 - (T/T_c)^{3/2}]. \quad (9)$$

Provenance: known ideal-gas statistics, proposed companion application. The adopted kinetic scale is G times enclosed monopole mass divided by $3r$; it is not a measured temperature. Fraction f of each original shell is moved to sr and the rest stays. A shared inner-bin fit gives $m=17.78$ eV/c squared and $s=0.8032$. These values are effective fit parameters. Massive bound states would have to differ from the original light-speed traveling excitations; thermalization and particle-number behavior are unresolved. Published superfluid models motivate investigating phases [4] but do not derive this recipe.

With companion self-energy U_d , external-potential energy U_b and necessary global support energy K , the cooling requirement is

$$E_{\text{mech}} = K + U_b + U_d, \quad Q_{\text{settle}} = E_{\text{mech},i} - E_{\text{mech},f}. \quad (10)$$

Provenance: standard energy accounting. U_d includes the self-energy factor one-half; U_b does not. K is assigned from the scalar force virial, a necessary global condition rather than a local equilibrium or stability proof. The tested shared model releases about $4.5e50$ J, or about $1.2e-8$ of deposited rest energy. A fully cooled solution requires this energy to leave the bound subsystem. Positive energy release alone does not determine the fitted stopping radius. Later coupled circular-endpoint calculations include redistribution of angular momentum to a 30-60 kpc receiving region. Their all-bin RMS changes from 6.34/10.91 to 6.28/8.10 km/s under the two baselines, with release $4.47e50/4.43e50$ J. These are endpoint constructions, not evolutionary or collective-stability proofs. Local pressure and saturation alternatives have not supplied a demonstrated joint solution.

4.2 Refraction and relativistic completion

$$\nabla \cdot [\varepsilon(\rho)\nabla\Phi] = 4\pi G(\rho_b + \rho_d). \quad (11)$$

Provenance: density-dependent gravitational permittivity is an existing refracted-gravity approach [5]. Our companion-density driver and retained deposition profile are adaptations. Tested variants do not improve reserved outer Milky Way predictions, so refraction remains a secondary experimental branch. A nonrelativistic Φ does not alone determine light bending: a conventional weak-field metric involves a second potential Ψ , and lensing probes their sum. No independent lensing multiplier is adopted.

4.3 Microscopic conversion and environmental time

Photon-graviton mixing in an external magnetic field is an established theoretical process [6], but its ordinary coupling is weak and stationary whole-photon conversion does not redshift surviving photons. A proposed inelastic interaction or time-dependent field must calculate frequency change, photon-number change, momentum transfer, angular scattering and timing together. QED-like and scalar/axion effective-field descriptions remain candidates; string-inspired fields are a source of possible interactions, not numerical predictions without specified couplings [7]. None currently derives equations (1), (5) and (7) together.

For an illustrative Poisson process removing fraction δ per event, mean event count Λ gives mean energy $E_0 \exp(-\Lambda \delta)$ and relative energy variance $\exp(\Lambda \delta^2) - 1$. These are known Poisson identities applied to a hypothesis. At the archived α and a 100-million-light-year path, an illustrative added relative linewidth of $1e-5$ requires per-event loss below $1.31e-8$ and at least roughly 582,000 events. This is a design example, not an observed exclusion limit.

4.4 Companion-powered nuclear emission

$$L_\gamma = f_\gamma P_{\text{release}}, \quad \dot{E}_{\text{res}} = P_{\text{capture}} - P_{\text{release}}. \quad (12)$$

Provenance: ordinary reservoir bookkeeping; attributing nuclear power to companions is our optional hypothesis. All released channels must be included. The radiated fraction f_γ is not the retention fraction η . Binding-energy release preserves constituent rest mass at Newtonian order; rest-energy evaporation changes the permanence postulate unless replenished. Observed black-hole-environment light is not established Hawking radiation, and no escape from inside a conventional horizon is assumed.

At a Sagittarius A* reference luminosity of $1e29$ W or less [8], pure photon output corresponds to about $1.77e-11$ solar masses/year of energy. The representative M87 core luminosity $2.7e35$ W [9] corresponds to $4.77e-5$ solar masses/year; its jet beaming, aperture and excluded mechanical power limit that comparison. These values reconstruct required throughput from observed brightness, rather than predicting brightness. Routing the whole Galactic settling budget through the faint Sgr A* reference channel would take about $1.43e14$ years. This ratio imposes no universe age: it instead demands an explicit routing, efficiency and activity-history model.

4.5 Transfer, angular scattering and rate consistency

Subsequent fixed-gap calculations separate the energy removed per interaction from the number of interactions. For independent small steps of energy Δ , the conditional relative energy variance is approximately $\Delta z/E_{\text{emit}}$ in the no-removal ladder. Maintaining an illustrative relative linewidth of $1e-5$ at $z=1$ for 2 eV input therefore requires Δ no larger than about $2e-10$ eV. Added photon-removal channels impose separate brightness constraints. These are design requirements under the stipulated stochastic process, not measured universal limits. The fractional-loss example above and the fixed-gap ladder are distinct mechanisms.

Finite spatial overlap can narrow scattering but also suppress its total rate. In the tested Gaussian heavy-store approximation, matching a scattering angle that decreases with apparent source size produces a rate proportional to inverse source-distance squared times $\exp[-(\text{distance}/\text{cutoff})^2]$. At fixed coupling and target density, the effect then saturates near the source. The favorable angular prescription cannot be combined with an unchanged long-path loss coefficient. A positive continuum of gaps alone also fails to flatten the earlier point-interaction color dependence; inverse-designed spectral responses remain hypotheses.

Source-observer geometry further distinguishes accumulated direction change from apparent image width. One fixed-angle optical example over 30.66 Mpc gives about 0.520 arcseconds apparent directional RMS and 5025 seconds mean geometric delay. A prescribed source-size-tracking kernel gives much smaller moments, but lacks a derived local cause and fails the unchanged-rate Gaussian closure. These are conditional moments, not image fits or gamma-ray timing predictions. A stationary delay distribution does not by itself provide redshift-proportional event dilation.

4.6 Reciprocal loading and finite protected storage

A candidate paired-mode interaction raises a store by Delta while transferring a quantum from energy E_H to $E_L=E_H-\Delta$. Its Hermitian conjugate permits reversal. In an incoherent, equally weighted mode-pair approximation, known bosonic factors give

$$A = \Gamma n_H(1 + n_L), \quad B = \Gamma n_L(1 + n_H), \quad r = \frac{B}{A}. \quad (13)$$

Provenance: standard ladder-operator factors applied to our proposed effective interaction. Forward preference requires $n_H > n_L$; smooth same-direction thermal occupations do not supply it. Maintained source and escape fields can sustain loading in the tested open-system closure, but their geometry, absolute coupling, recoil and connection to traveling companions are not derived. For a one-step optical route, a 2 eV input contributing 1e-8 eV processes 2e8 times its net assembly power, with most energy remaining in outgoing light rather than heat.

Normalized collective states do not provide a free N-squared first-excitation rate from an empty domain: the factor is N. Partly excited symmetric states can have stronger enhancement, but require preparation energy and enhance reverse transitions too. These are known collective-spin results [12], not new graviton physics. Repeated assembly can succeed despite a small probability per attempt, while requiring large recycled energy traffic; the representative unbiased ladder circulates about 200,000 times its net assembly energy.

Adding one finite protected level and a relaxation mode with occupation n gives, in the zero-protected-decay equilibrium of the tested model,

$$P_{\text{protected}} = \frac{n + 1}{n + 1 + rn}. \quad (14)$$

Provenance: a derived stationary result of our specified finite-state model. Collective size cancels from this occupancy, although it affects replenishment rates. Fifty-four stationary cases include reverse export and conserve energy. Nonzero leakage requires replenishment; zero leakage is a stipulated lifetime, not an explanation of permanence. One finite level also cannot accumulate unlimited energy.

When the full fitted Milky Way energy inventory is assigned to tiny single-excitation sites, the illustrative site/mode ratios become extremely large. A particular homogeneous escape closure requires protected lifetimes above roughly 1e32-1e41 years for its 120 kpc, 90%-protected examples. These are conditional required lifetimes, not cosmic-age constraints. Larger collective release quanta can ease mode crowding, but require a consistent assembly, reverse-transfer and protection mechanism. Thermal and nonthermal outgoing channels remain separate alternatives; energy in transit is not deposited energy.

4.7 A threshold interpretation of the one-third reference

The reference response has an exact representation as a capacity-weighted mixture of ordinary saturation laws. With positive threshold t and exponent q ,

$$\rho_q(t) = \frac{\sin(\pi q)}{\pi} \frac{t^{q-1}}{1 + 2t^q \cos(\pi q) + t^{2q}}. \quad (15)$$

$$\int_0^\infty \frac{X}{X+t} \rho_q(t) dt = \frac{X^q}{1+X^q}, \quad q = \frac{1}{3}. \quad (16)$$

Provenance: known positive fractional-response mixture mathematics [13]. Selecting this distribution from the desired curve is inverse design, not a first-principles derivation of one-third. Equilibrium occupied capacity is not automatically permanent capture efficiency. A physical model must derive the threshold weighting and connect it to the incident field and stored mass.

Truncating and renormalizing this distribution to thresholds from $1e-6$ to $1e6$ changes predicted rotation speeds by at most 0.640 km/s across the existing 149 galaxies without refitting. A narrower $1e-3$ to $1e3$ range changes some speeds by 7.03 km/s and worsens aggregate validation/test velocity RMSE. Broad cutoffs preserve the reference by construction; they neither establish the distribution's existence nor improve its main residuals.

4.8 Formation history and retention after fading

One explicit local kinetics is $df_t/dt = \lambda[X(1-f_t) - t f_t]$. From empty storage under constant X , its standard solution is

$$f_t(u) = \frac{X}{X+t} [1 - e^{-(X+t)u}], \quad u = \lambda \Delta t. \quad (17)$$

Provenance: known linear rate-equation solution; applying a shared λ and the selected threshold distribution to companions is hypothetical. The overall time scale is undetermined. Across the catalogue inputs, reaching 90% equilibrium takes dimensionless durations from 0.08478 to 102.462 . Shared-duration scans produce mixed score changes and do not select a new preferred law.

After individually stopping the source at that target, losing the first 10% of formed storage takes 4.92%-41.79% of the formation time. This ratio is independent of λ and uses no universe age. A slower residual tail persists, but does not make the whole reservoir permanent. These controlled histories are not observations of the named galaxies and do not model continued distant illumination. Equilibrium requires balancing capture and release; a successful amplitude fit does not establish the required energy throughput.

5. Executed results and limits of the evidence

The project record contains both successes and failures. Parameter selection and previously inspected observations must not be described as blind tests. Reported RMS values are descriptive, not likelihood-based evidence for the entire theory.

Test	Current result	What it establishes
SPARC rotation, 149 galaxies	Companion validation/test RMS 32.49/23.59 km/s; MOND 26.88/16.40	Companions improve the ordinary-only control, but tested MOND is better
Six Coma shear bins	Companion/NFW omitted-bin sums 6.86/6.82	Comparable freely fitted shapes; no transferred supply normalization
Six lens galaxies	About 14% lens-angle RMS	A material unresolved motion/lensing discrepancy
Earlier imposed Milky Way settling	Inner RMS 3.68/11.83 becomes 2.74/6.70 km/s	A shared conditional inner-fit improvement under two matter baselines
Earlier pilot outer bins	RMS remains 8.36/9.78 km/s	No outer improvement from the selected central rearrangement
Refraction sweep	No selected outer improvement	No reason to promote the tested refraction family
Coupled Milky Way endpoints, 30-60 kpc receiver	All-bin RMS 6.28/8.10 km/s	Conditional angular-momentum and energy accounting, not stability
Finite thresholds, 149 galaxies	12-decade range shifts speeds at most 0.640 km/s	Preserves reference approximately; no improved main residuals
Threshold histories, 149 intensity inputs	First 10% loss takes 4.92%-41.79% of formation time	Conditional kinetics, not observed galaxy ages
Reciprocal protected reservoir	54 stationary cases; balanced energy flows	Replenishment can improve; permanence remains assumed
Nuclear release	Energy-throughput estimates only	A possible diagnostic channel, not a predicted spectrum or luminosity

SPARC inputs [10] have 89 training, 29 validation and 31 test galaxies in the project split; all have been exposed during development. Companions have three shared fitted constants versus one for the tested MOND law. Restricted shared NFW mappings and per-galaxy NFW fits use different information budgets and cannot be pooled into one ranking. The Milky Way phase pilot uses a spherical gas monopole and two archived

stellar baselines, with 20 inner bins for selection and 18 outer bins for comparison. It is not a comparison of two independent galaxies or individual bulge-star orbital predictions [11].

Earlier wave/particle support, saturation, inward migration and coupled exchange tracks remain in the research record. None supplies a demonstrated joint solution for stability, absolute supply and lensing. Numerical mass/energy checks and refinement protect against implementation errors; they do not verify the physical assumptions. The accompanying cross-scale-performance.md and linked executable reports contain the detailed audit trail.

6. Evidence that would materially strengthen the theory

The following outcomes are the main work program. Each explains what is needed and why failure matters. They are not claims already achieved.

1. A common mechanism with fewer adjustable choices. Specify an interaction and local state equations that yield conversion, capture and support with shared constants. Derive the one-third exponent or replace it with a better justified prediction. Otherwise independent fitting rules can conceal incompatible physics.
2. Absolute supply from independently constrained radiation. Integrate source luminosities, travel, capture, retention and release, keeping age/history as explicit inferred parameters rather than choosing unlimited history to rescue a deficit. Predict effective stored gravity without fitting C_0 separately to the needed mass. Otherwise we have renamed a missing-mass profile rather than explained its origin.
3. Redshift and time stretching from the same propagation law. Use distances independent of redshift, control velocities, predict the same fractional spectral shift across tested bands, and reproduce event durations, sharp images and brightness. Include GW/EM timing. Otherwise matching a distance-redshift curve alone does not explain the observed propagation phenomena.
4. Joint motion and lensing prediction. Use one relativistic response and one deposition profile to predict circular speeds, stellar velocity distributions and strong/weak lensing in the same objects. Include distances, gas, mass-to-light and orbit uncertainty. Otherwise a rotation success cannot repair a lensing failure through an unrelated correction.
5. Genuine transfer to untouched systems. Freeze parameters and analysis choices, then predict additional galaxies and clusters spanning surface brightness, gas content, morphology and environment. Compare residuals, scatter and parameter burden against properly specified MOND and dark-halo models with identical inputs. Otherwise in-sample improvement may reflect flexibility or selection.
6. Formation, stopping and persistence. Solve evolving density, pressure or orbital support, angular momentum and cooling, including feedback from the companion field. Check Solar System and binary constraints in the appropriate limits. Otherwise a useful profile may never form, may collapse, or may disrupt systems already explained well.
7. A radiation prediction distinct from ordinary sources. From the same fixed settling law, predict where energy emerges, its spectrum and variability before using measured brightness. Test both diffuse emission and the optional nuclear channel, including nondetections and jet mechanical power. Otherwise any observed glow can be assigned to companions after the fact.

8. Broader nonexpanding-universe consistency. Address the measured microwave spectrum and angular correlations, large-scale clustering, abundances and thermal histories without importing an expanding-model solution as a premise. Retain observations even when their conventional explanation is set aside. Otherwise a gravitational phenomenology cannot claim to replace a cosmological framework.

7. What would justify a claim of advantage?

A restricted claim is the first realistic target: for example, a frozen companion law predicts joint motions and lensing in a new sample more accurately than specified competitors with comparable or lower effective flexibility. That would be a meaningful result even before every cosmic observable is explained. It should be reported with measurement covariance, nuisance-parameter uncertainty, model complexity and all failed targets included.

A stronger advantage would come from an independently confirmed connection between lost photon energy, accumulated gravity and released radiation. An environmental dependence or radiation feature predicted in advance, absent from the tested competing models, would be especially valuable. The signature must be quantitatively calculated; an unexplained residual is not automatically companion evidence.

Directly detecting an interaction or state associated with companions would be compelling, but is not the sole route to a viable theory. Conversely, dark-matter direct-detection limits do not automatically promote this theory. Both frameworks must meet the evidence appropriate to their claims. Since companions add an unseen gravitational component, the distinct scientific claim is its radiation-derived origin and calculable dynamics, not simply the absence of unseen matter.

8. Continuing work across tracks

The reference track remains exact-third deposition with transparent input and parameter accounting. The settling track seeks local support and cooling equations that determine the contraction rather than fitting it. The propagation track seeks an inelastic or environmental-time interaction that predicts both spectra and event timing. The gravity track seeks one relativistic lensing/motion response; unsuccessful refraction variants remain documented. The emission track now includes companion-powered nuclear radiation as an optional branch while retaining diffuse and invisible outgoing channels. The data-comparison track freezes rules before applying them to additional systems.

These tracks must eventually share one energy and momentum ledger. Until then they are labeled alternatives or partial closures; results from incompatible versions must not be assembled into a fictitious single successful theory. No background run or completed physical goal is implied by retaining a track in this work program.

9. Conclusion

The reference remains an empirical radiation-conditioned extra-gravity model with substantial unresolved physics. New work narrows the permissible mechanisms: angular narrowing must be coupled to its rate cost; collective enhancement must include normalized states and reverse flow; storage must include finite capacity and energy release. A threshold mixture interprets the exact-third curve but does not derive its exponent, and its simplest kinetics does not make deposits permanent. Existing galaxy sensitivities are not fresh blind evidence. The next advance must connect an independently specified interaction and source history to a joint gravitational and radiative prediction. This paper does not establish a replacement for dark matter, MOND or expanding cosmology.

References

- [1] LZ Collaboration / Berkeley Lab (2025). LZ Sets a World's Best in the Hunt for Galactic Dark Matter. [Experiment result and search scope](#).
- [2] DES Collaboration (2024). Slow supernovae show cosmological time dilation out to z approximately 1. [arXiv:2406.05050](#). Used as an observed timing constraint; expansion is not adopted as this paper's premise.
- [3] LIGO/Virgo and partner collaborations (2017). Gravitational Waves and Gamma-Rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A. [Publication record](#).
- [4] Berezhiani, Famaey and Khoury (2018). Phenomenological consequences of superfluid dark matter with baryon-phonon coupling. [arXiv:1711.05748](#).
- [5] Sanna, Matsakos and Diaferio (2023). Covariant formulation of refracted gravity. [arXiv:2109.11217](#).
- [6] Palessandro and Rothman (2023). A Simple Derivation of the Gertsenshtein Effect. [arXiv:2301.02072](#).
- [7] Cicoli, Goodsell and Ringwald (2012). The type IIB string axiverse and its low-energy phenomenology. [arXiv:1206.0819](#).
- [8] Event Horizon Telescope Collaboration (2022). First Sagittarius A* Results I. [doi:10.3847/2041-8213/ac6674](#).
- [9] Prieto et al. (2016). The central parsecs of M87: jet emission and an elusive accretion disc. [MNRAS 457, 3801](#).
- [10] Lelli, McGaugh and Schombert; SPARC data and related publications. [Project publications](#). Project-specific splits and fits are documented in the repository.
- [11] Eilers et al. (2019). The Circular Velocity Curve of the Milky Way from 5 to 25 kpc. [arXiv:1810.09466](#).
- [12] Dicke, R. H. (1954). Coherence in Spontaneous Radiation Processes. [doi:10.1103/PhysRev.93.99](#). Used for established collective-state physics, not evidence for companions.
- [13] Tuncer, E. (2010). Geometrical Description in Binary Composites and Spectral Density Representation. [Materials 3, 585](#). Used for known spectral-mixture mathematics.

Reproducibility and research status

This revision summarizes evidence through main revision 6e6b022, updating the earlier v1.0 snapshot at 35a7ea7. The v1.0 source and PDF remain recoverable in Git history. Detailed protocols, formulas, results and provenance are linked in `cross-scale-performance.md` and `research_work/results/companion-extensions/`. Key new executable records include coupled-torque, escape-feedback, site-mode-budget, collective-assembly, source-outlet, observer-transport, kernel-rate-closure, collective-normalization, collective-reservoir, threshold-mixture, threshold-galaxies, threshold-history and threshold-memory, each with its report and result JSON. The numerical tables use their recorded outputs; no new fitting is performed for this manuscript revision. The observed-data sample contains 149 galaxies and 3150 accepted rotation measurements in already-exposed partitions. Numerical verification validates stated calculations, not their assumptions. Computational assistance was used in analysis and drafting; independent scientific review remains necessary.